Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Abstract

We addressed two long-standing gaps in synthetic equity path generation: single-asset stylized facts (heavy tails, regime persistence, volatility clustering) and multi-asset factor structure (cross-sectional dependence with marginal preservation). We developed a hybrid hidden Markov framework that discretized excess growth rates into Laplace quantile-defined states and augmented regime switching with a Poisson jump-duration mechanism, enforcing realistic tail-state dwell times without Baum-Welch optimization. We then composed the per-asset hybrid HMM marginals into a multi-asset Single-Index Model via a closed-form variance correction that rescaled the idiosyncratic draw to preserve the marginal variance while recovering the calibrated factor loading; a sign-preserving clip and an R^2-driven branch handled high-leverage and index-tracking edge cases.

Evaluated on a 424-ticker US equity and ETF universe against the 2014–2024 SPY history, the hybrid HMM marginals passed a per-asset Kolmogorov–Smirnov test against the real univariate distribution at 97.6% in-sample and 94.4% out-of-sample, outperforming Bootstrap, Gaussian, Laplace, GARCH, and GRU baselines on combined distributional and tail metrics. The variance-corrected SIM composition recovered the factor loading to a median absolute error of 0.018 and matched the leading residual-fit alternatives on a per-ticker one-day 99% Value-at-Risk backtest (mean exceedance 0.95% versus the nominal 1%, Kupiec pass rate 80%). We documented one downstream caveat: an iid-residual diversification-return artifact under daily rebalancing, addressable via a uniform location-shift correction anchored to a documented long-run prior. All code and per-asset fitted marginals are released as part of `JumpHMM.jl`.

Keywords: hidden Markov model; jump-diffusion; single-index model; synthetic financial data; variance correction; Value-at-Risk backtest; volatility clustering; heavy-tailed marginals.

1 Introduction

Portfolio construction, risk attribution, and stress testing often take long synthetic time series of asset growth rates as inputs. A good generator has to satisfy two things the downstream consumer cares about. First, each single asset's synthetic series should look like its real counterpart when you histogram it day by day: the tails should be as heavy as real tails, the kurtosis should match, regimes and jumps should show up with roughly the right frequency; in statistical language this is the *marginal distribution* of the asset. Second, if we stack synthetic series across many assets they should co-move the way real markets co-move: on a day when the market plunges, stocks with high factor loadings should plunge with it, tail co-movement should look realistic, and factor structure should be intact; this is the *cross-sectional dependence* across assets. These two requirements are largely orthogonal, and off-the-shelf generators tend to handle only one of them. Per-asset generators built from hidden Markov models [1], generalized autoregressive conditional heteroskedasticity (GARCH) families [2], or empirical bootstraps [3] handle the first: fit one of these models to AAPL's growth-rate history and its synthetic paths carry AAPL's tail thickness, kurtosis, and regime structure. But the fit knows nothing about market factors, e.g., SPY, or any other assets, e.g., MSFT. Stacking the per-asset draws therefore produces one synthetic growth-rate path per asset, each matching its own asset's full growth-rate marginal. But the paths carry no joint structure: on a day when synthetic SPY plunges, synthetic AAPL has no reason to plunge with it (but real AAPL would).

The standard recipe for adding factor structure is a linear factor model, either the one-factor single-index model (SIM) of Sharpe [4] or a multi-factor variant such as Ross's arbitrage pricing theory [5] or Fama-French [6]. We develop the construction for the one-factor SIM, which writes each asset's growth rate as an intercept plus a market-loaded term plus an idiosyncratic residual; the variance correction we propose extends to the multi-factor case without structural change. Applied naively, the recipe plugs a synthetic full growth-rate path from the per-asset generator (fit to the asset's real growth-rate history and matching its full growth-rate marginal) straight into the SIM as the idiosyncratic residual. Because the full growth rate already contains the market-driven part, using it as the residual double-counts the market variance and inflates the marginal variance of the composed series by an amount that grows quadratically in the factor loading. For high-loading assets the inflation is large enough to be visible in the empirical marginal: the body widens, the kurtosis drops, and the heavy tails that the generator was built to deliver are pushed out into a Gaussian-flavored distribution driven by the market term, defeating the purpose of fitting a heavy-tailed generator in the first place. The opposite failure mode, drawing the residual from a Gaussian with variance taken from the real ordinary least squares (OLS) fit, recovers the SIM coefficients exactly but erases the marginal structure entirely. Neither naive option satisfies both targets at once, and closing that gap is the problem the rest of this paper addresses.

This paper describes a closed-form variance correction that sits between the two failure modes. The construction rescales the per-asset generator draw by a scalar chosen so that the variance algebra works out to preserve the generator marginal; the scalar admits a closed form whenever the market component accounts for less than the full generator variance. When that condition fails (high-loading assets paired with a low-volatility generator, or a synthetic market more volatile than the calibration market), the factor loading is clipped downward against a floor that reserves a configurable share of the marginal variance for idiosyncratic noise, with a sign-preserving clip that keeps defensive assets defensive. A separate branch handles index-tracking assets, where the calibrated coefficient of determination (the R^2 from OLS of the asset's real growth-rate history against the market path) rather than the generator marginal is the quantity worth preserving; SPY against itself recovers as the degenerate perfect-tracking limit with no special-casing. Branch selection is driven entirely by the calibrated R^2, and each asset carries a persisted construction flag (`hybrid`, `hybrid-clipped`, `r2-preserve`) so downstream consumers can audit which path each asset took, and the procedure generalizes to any new ticker without external metadata.

In this study, the distinction between the present paper and the companion `JumpHMM.jl` validation paper of Alswaidan and Varner [7] is worth stating explicitly: the companion fits and validates the per-asset JumpHMM marginals themselves and asks whether the generator reproduces real marginal and joint statistics when used as-is, while the present paper takes the generator as given and asks how it should be composed with a single-index market factor without breaking either target. The variance correction, the sign-preserving clip rule, the R^2-preserving branch, the construction flag, and the empirical comparison of composition rules on a common universe are specific to this work; the two contributions are orthogonal. We tested the construction on the 424-ticker US equity universe and SPY market path used by the companion paper, comparing six composition recipes on the same generators and the same market path: the naive composition, a Gaussian-residual SIM that recovers the calibrated factor coefficients and R^2 but loses the marginal structure, the hybrid construction proposed here, a

JumpHMM marginal fit directly to the OLS residual series (composed with no variance correction), a Politis–Romano stationary bootstrap of the real residual history [3], and a per-ticker GARCH(1,1)-t fit to residuals [2]. We measured SIM recovery, marginal fidelity, and per-ticker one-day Value-at-Risk accuracy, reporting each per construction-flag branch. All six composers recovered the factor loading to a median absolute error of 0.018 (hybrid, residual-fit, bootstrap, GARCH) or 0.022 (naive) (Table 1). On marginal fidelity the hybrid construction held the Kolmogorov–Smirnov pass rate at 50%, against 4% for the naive composition and below 1% for the Gaussian baseline; the three residual-fit alternatives attained pass rates of 67 to 76% (Table 1). On the 99% Value-at-Risk backtest, hybrid and JumpHMM-on-residuals delivered statistically indistinguishable coverage (Kupiec pass rates 80% and 83% respectively), while the strawman composers failed coverage at around 45% (Table 5). These results indicated that the marginal-versus-factor tradeoff long assumed in SIM composition was not fundamental: a single closed-form scalar correction closed it to within a few percentage points of a dedicated residual-fit generator (JumpHMM-on-residuals, block bootstrap, or GARCH(1,1)-t) on marginal-body fidelity, and matched the residual-fit generator on a one-day 99% Value-at-Risk backtest against the real growth-rate histories. The hybrid composition rule traded a small body-fidelity gap for generator agnosticism: any per-asset full-return marginal, including heavy-tailed HMM-with-jumps or regime-switching generators calibrated once for other purposes, could enter the factor pipeline unchanged. The correction generalizes to richer factor structures (multi-factor SIM, lagged-factor SIM) without changing the underlying algebra.

2 Method

We work throughout in continuously compounded annualized growth rates, $g(t) = [\log p(t) - \log p(t-1)]/\Delta t$, where $p(t)$ is the closing price on trading day t and Δt is the fraction of a trading year spanned by a single trading day (we use $\Delta t = 1/252$ in the experiments below, so the per-day log return is $g(t)\,\Delta t$). Working in annualized growth rates rather than raw log returns keeps variances and covariances in the units customary in risk reporting. Growth rates compose linearly under aggregation in time and are the natural unit for variance and covariance arithmetic, which the construction below relies on heavily. All variances and covariances refer to sample moments computed over the relevant time window unless stated otherwise. The method composes a synthetic asset growth-rate path of the form:

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T, \tag{1}$$

from an independent generator draw $\tilde{\varepsilon}_i \in \mathbb{R}^T$ and a shared market growth-rate path $g_m \in \mathbb{R}^T$ over T trading days (formally introduced below, with $\tilde{\varepsilon}_i$ assumed independent of g_m and $\mathbb{E}[\tilde{\varepsilon}_i] = 0$) by rescaling the generator draw as $\varepsilon_i = s_i\,\tilde{\varepsilon}_i$ with a scalar $s_i \in (0, 1]$ chosen to preserve one of two per-asset marginal targets while recovering the calibrated single-index model (SIM) coefficients (α_i, β_i) in the large-T limit.

2.1 Problem setup

We are given a single market growth-rate path $g_m \in \mathbb{R}^T$ over T trading days, with sample mean μ_m and sample variance $\sigma_m^2 = \mathrm{Var}[g_m]$. In our experiments the market path is the SPY total-return growth-rate series; the construction is agnostic to whether g_m is a single-name proxy, a synthetic index, or a model-generated path.

For each of N assets indexed by $i = 1, \ldots, N$, we are given a draw $\tilde{\varepsilon}_i \in \mathbb{R}^T$ from a generator that produces growth rates with the target marginal distribution: heavy tails, regime structure, jumps, or whatever the application requires. Write $\sigma_{\mathrm{gen},i}^2 = \mathrm{Var}[\tilde{\varepsilon}_i]$ for the sample variance of the draw. The construction is agnostic to how the generator is fit; in our experiments we use the per-asset hidden Markov model with Student-t emissions and explicit jump components from `JumpHMM.jl` v0.1.0 (commit `9be8099`)[1], fit to the asset's own full growth-rate history (not to residuals after market detrending) using the package-default EM tolerance and maximum-iteration cap. Because the generator's variance reflects the full return rather than the idiosyncratic part, the rescaling $\varepsilon_i = s_i\,\tilde{\varepsilon}_i$ with $s_i^2 = 1 - \beta_i^2\sigma_m^2/\sigma_{\mathrm{gen},i}^2$ is exactly what prevents the market term in the SIM composition from double-counting variance that is already carried by $\tilde{\varepsilon}_i$. The generator draws are demeaned, $\mathbb{E}[\tilde{\varepsilon}_i] = 0$, so that the constant term in the SIM is carried entirely by α_i. We assume $\mathrm{Cov}[\tilde{\varepsilon}_i, g_m] \approx 0$ for each i, which holds when the generator and market path are independent draws (the typical use case),

[1] `https://github.com/varnerlab/JumpHMM.jl`

and holds after a copula rank reorder when the copula is sampled independently of g_m (verified empirically across the 423-ticker universe in Fig. S1). If the generator already carries non-trivial market correlation, the residual covariance can be subtracted from β_i before applying the correction; we return to this point in the discussion.

For each asset we are given a calibrated triple $(\alpha_i, \beta_i, R^2_{i,\text{real}})$, estimated by ordinary least squares on the real return history of asset i against the real market path. These are the targets we want the synthetic construction to recover. The intercept α_i and slope β_i are the standard SIM coefficients; $R^2_{i,\text{real}}$ is the real-data coefficient of determination, used by the branch-selection step defined below.

2.2 Preservation criteria

We want the synthetic composition $g_i = \alpha_i + \beta_i^{\text{eff}}\, g_m + \varepsilon_i$, with effective slope β_i^{eff} (equal to β_i on the unclipped branch and the attenuated value of Eq. (6) otherwise), to satisfy three properties in the large-T limit. The first is *SIM recovery*: ordinary least squares of g_i on g_m returns the calibrated intercept and slope, $\hat{\alpha}_i = \alpha_i$ and $\hat{\beta}_i = \beta_i^{\text{eff}}$. On the unclipped branch, $\beta_i^{\text{eff}} = \beta_i$ and the OLS output reproduces the calibration exactly; on the clipped branch, β_i^{eff} is the attenuated value defined by Eq. (6), and the attenuation is persisted alongside the output so it is auditable. The second is *marginal fidelity*: each asset is assigned one of two per-asset targets based on its calibrated R^2, not both at once. Assets whose generator marginal (heavy tails, regime structure, jumps) is the quantity of interest receive the variance-preserving target $\text{Var}[g_i] = \sigma^2_{\text{gen},i}$; assets that track the market closely receive the R^2-preserving target $R^2 = R^2_{i,\text{real}}$, measured by the same OLS regression as above. The branch chosen for each asset is recorded as a construction flag so downstream consumers can audit which target was pursued. The third is *cross-sectional dependence*: any joint dependence structure applied to the residuals $\{\varepsilon_i\}_{i=1}^N$ passes through unchanged to the composed series $\{g_i\}_{i=1}^N$, automatic given the linearity of Eq. (1) in ε_i. The joint structure itself (empirical copula, factor copula, t-copula) is supplied as a separate design choice, and we report cross-sectional fidelity for the empirical-copula choice in the results below.

The naive composition $g_i = \alpha_i + \beta_i\, g_m + \tilde{\varepsilon}_i$ satisfies SIM recovery and cross-sectional dependence but violates marginal fidelity: its marginal variance is $\sigma^2_{\text{gen},i} + \beta_i^2 \sigma_m^2$, an inflation that grows quadratically in β_i. The rest of this section fixes this inflation with a closed-form scalar correction, handles the degenerate high-β case with a sign-preserving clip, treats index-tracking assets through an R^2-preserving branch, and ties the pieces together in an algorithm that also states the resulting properties of the composed series.

2.3 Variance correction

Form the candidate residual $\varepsilon_i = s_i\, \tilde{\varepsilon}_i$ for some scalar $s_i \in (0, 1]$, and require that the composed series have the same variance as the generator,

$$\text{Var}[\beta_i\, g_m + \varepsilon_i] \;=\; \sigma^2_{\text{gen},i}. \tag{2}$$

Under the independence assumption $\text{Cov}[\tilde{\varepsilon}_i, g_m] \approx 0$, the cross term in (2) vanishes and

$$\beta_i^2\, \sigma_m^2 \;+\; s_i^2\, \sigma^2_{\text{gen},i} \;=\; \sigma^2_{\text{gen},i}, \tag{3}$$

giving

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\, \sigma_m^2}{\sigma^2_{\text{gen},i}}} \tag{4}$$

Define the *market loading ratio*

$$\rho_i \;\equiv\; \frac{\beta_i^2\, \sigma_m^2}{\sigma^2_{\text{gen},i}}, \tag{5}$$

the share of the generator variance consumed by the market component. Then $s_i^2 = 1 - \rho_i$, valid whenever $\rho_i < 1$. The naive composition $g_i = \alpha_i + \beta_i g_m + \tilde{\varepsilon}_i$ corresponds to $s_i = 1$ and inflates $\text{Var}[g_i]$ by exactly $\beta_i^2 \sigma_m^2$.

2.4 Beta clipping with an idiosyncratic floor

Equation (4) breaks down when $\rho_i \geq 1$: the market loading alone would account for more than the entire generator variance, leaving no room for idiosyncratic noise. This case arises for high-β_i assets paired with low-volatility generators, or whenever the synthetic market g_m is more volatile than the market on which β_i was calibrated.

To handle it, impose a floor $f \in (0, 1)$ on the idiosyncratic share by requiring $s_i^2 \geq f$, and clip β_i downward whenever the floor would be violated. Writing the clip threshold as $\rho_i^{\max} = 1 - f$,

$$\beta_i^{\text{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\ \operatorname{sign}(\beta_i)\sqrt{(1-f)\,\dfrac{\sigma^2_{\text{gen},i}}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1-\rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{6}$$

Sign preservation in (6) keeps defensive (negative-β) assets defensive even when clipped. The floor f is a tunable knob; in the experiments below we use $f = 0.10$, which guarantees that at least 10% of each asset's variance comes from idiosyncratic noise. When β_i is clipped we log a warning and persist both β_i and β_i^{eff} so downstream consumers can audit the attenuation.

2.5 R^2-preserving branch for tracker assets

The variance-preserving construction targets $\operatorname{Var}[g_i] = \sigma^2_{\text{gen},i}$. That is the right target when the generator marginal carries information one wants to keep. For some assets the generator marginal is not really the quantity of interest; the SIM relationship to the market is. Index ETFs are the canonical example. SPY against itself has $\beta = 1$, $R^2 = 1$, $\sigma_\varepsilon = 0$ exactly. QQQ against SPY has $R^2 \approx 0.84$, SPYG against SPY has $R^2 \approx 0.86$. For these tickers we require $\hat{\beta}$ and R^2 to recover their calibrated values, neither inflated to one by dropping ε nor implied by unrelated marginal-variance bookkeeping.

We branch on the real-data calibration $R^2_{i,\text{real}}$. Pick a threshold R^2_{preserve} that separates index-tracking assets from genuine single stocks; in our universe the cumulative distribution sorts itself cleanly, with index ETFs above 0.80 and the most market-like single names below 0.65 (Fig. 7), so $R^2_{\text{preserve}} = 0.80$ captures the trackers without sweeping in any genuinely idiosyncratic name. For any asset with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$, the population R^2 identity for (1),

$$R^2 \;=\; \frac{\beta_i^2\,\sigma_m^2}{\beta_i^2\,\sigma_m^2 + \sigma^2_{\varepsilon,i}}, \tag{7}$$

inverts to give the residual variance that hits the calibrated R^2:

$$\boxed{\sigma^2_{\varepsilon,\text{target}} \;=\; \beta_i^2\,\sigma_m^2 \cdot \frac{1 - R^2_{i,\text{real}}}{R^2_{i,\text{real}}}} \tag{8}$$

Draw $\tilde{\varepsilon}_i$ from the ticker's generator and rescale,

$$\varepsilon_i \;=\; \sqrt{\frac{\sigma^2_{\varepsilon,\text{target}}}{\sigma^2_{\text{gen},i}}}\,\tilde{\varepsilon}_i, \tag{9}$$

then copula-reorder and compose as in (1). The construction recovers both calibrated quantities by design: $\hat{\beta} \to \beta_i$ in the large-T limit because ε_i is uncorrelated with g_m after the rank reorder, and $R^2 \to R^2_{i,\text{real}}$ by construction.

When $R^2_{i,\text{real}} = 1$ exactly (SPY against itself), (8) gives $\sigma^2_{\varepsilon,\text{target}} = 0$, so $\varepsilon_i \equiv 0$ and $g_i = \alpha_i + \beta_i g_m$ deterministically. No special-casing is needed: the SPY identity falls out of the same expression with $\beta = 1$, $\alpha = 0$, $R^2 = 1$. If the rescale factor in (9) exceeds one, the generator's tails are being stretched to fit a target larger than the original draw; this happens when the synthetic market g_m has lower variance than the calibration market. We detect and flag this case so the loss of marginal fidelity is auditable.

Algorithm 1 Hybrid SIM composition with variance correction.

Require: Market path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2; per-asset generator draws $\tilde{\varepsilon}_i \in \mathbb{R}^T$ for $i = 1, \ldots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\text{preserve}} \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

1: **for** each asset $i = 1, \ldots, N$ **do**
2:     Compute generator variance $\sigma^2_{\text{gen},i} \leftarrow \text{Var}[\tilde{\varepsilon}_i]$.
3:     **if** $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ **then**
4:         $\sigma^2_{\varepsilon,\text{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}})/R^2_{i,\text{real}}$ {Eq. (8)}
5:         $\varepsilon_i \leftarrow \sqrt{\sigma^2_{\varepsilon,\text{target}}/\sigma^2_{\text{gen},i}}\,\tilde{\varepsilon}_i$
6:         $\beta_i^{\text{eff}} \leftarrow \beta_i$; flag $\leftarrow$ `r2-preserve`
7:     **else**
8:         $\rho_i \leftarrow \beta_i^2 \sigma_m^2/\sigma^2_{\text{gen},i}$ {Eq. (5)}
9:         **if** $\rho_i \leq 1 - f$ **then**
10:             $s_i^2 \leftarrow 1 - \rho_i$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; flag $\leftarrow$ `hybrid`
11:         **else**
12:             $s_i^2 \leftarrow f$; $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i)\sqrt{(1-f)\,\sigma^2_{\text{gen},i}/\sigma_m^2}$; flag $\leftarrow$ `hybrid-clipped`
13:         **end if**
14:         $\varepsilon_i \leftarrow s_i\,\tilde{\varepsilon}_i$
15:     **end if**
16:     *(Optional)* apply cross-sectional rank reorder to ε_i.
17:     $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}}\,g_m + \varepsilon_i$ {Eq. (1)}
18: **end for**

2.6 Branch selection, algorithm, and properties

The pieces combine into a single procedure (Algorithm 1). For each asset the calibrated $R^2_{i,\text{real}}$ selects a branch: assets with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ go through the R^2-preserving construction of Eq. (8), and the rest go through the variance-preserving construction of Eq. (4) with the clip of Eq. (6) engaged when the market loading ratio ρ_i would otherwise exceed $1 - f$. Each asset is tagged with a construction flag `r2-preserve`, `hybrid`, or `hybrid-clipped` that is persisted alongside the output, so downstream consumers can audit which path each asset took. The optional rank reorder in line 14 accepts any joint dependence structure (empirical copula, t-copula, factor copula); reordering preserves the marginal variance of ε_i, so the variance algebra of Eq. (4) is unaffected and whatever cross-sectional structure is applied to ε_i passes through to g_i unchanged.

By construction the composed series g_i satisfies the three preservation criteria above (SIM recovery, marginal fidelity, cross-sectional dependence) in the large-T limit. OLS of g_i on g_m recovers $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\text{eff}}$, with $\beta_i^{\text{eff}} = \beta_i$ unclipped and β_i^{eff} persisted alongside β_i when the clip triggers. Marginal variance matches $\sigma^2_{\text{gen},i}$ on the variance-preserving branch, either exactly, $\text{Var}[g_i] = \beta_i^2\sigma_m^2 + s_i^2\sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$, or by the choice of β_i^{eff} in (6) under clipping; on the R^2-preserving branch the marginal variance instead matches $\beta_i^2\sigma_m^2/R^2_{i,\text{real}}$, the value implied by the calibrated $(\beta_i, R^2_{i,\text{real}})$ rather than by the generator marginal. Tail behavior on the variance-preserving branch follows from the floor: because at least $f\,\sigma^2_{\text{gen},i}$ of the variance is reserved for ε_i, the tails inherited from $\tilde{\varepsilon}_i$ dominate whenever $(\beta_i^{\text{eff}})^2\sigma_m^2$ is small relative to those tails, while as β_i grows the marginal of g_i tightens toward Gaussian along the market-driven axis, mildly for low-β assets and noticeably at high β. The construction injects *contemporaneous* market dependence only; lagged dependence (lead-lag effects, market-driven volatility clustering) requires a richer construction, which we return to in the discussion. The pipeline (`Pipeline.jl`, in the released code) is modest in compute: fitting the 423 per-asset JumpHMM marginals runs once offline against the 2014–2024 price histories and is cached, while GARCH(1,1)-t residual paths are simulated per (ticker, replication) at scoring time rather than cached, so each run consumes a fresh draw. The scoring loop across $423 \times 100 = 42{,}300$ per-ticker replications per composer runs in minutes on a 2023 MacBook Pro

(Apple M2 Pro, 16 GB RAM), and the full six-composer sweep reported below completes within an hour on the same hardware.

3 Results

We evaluated the three composers on a fixed set of per-asset JumpHMM marginals calibrated against the real SPY growth-rate path. This isolated the effect of the composition rule from the effect of market-path randomness: all three composers consumed the same real SPY history and the same per-ticker innovation draw, so metric differences across composers reflected the composition rule alone. The ticker universe was a 424-asset US equity/ETF set drawn from Polygon.io daily-close histories, filtered to tickers with a complete in-sample trading history matching the reference length of AAPL ($T = 2{,}767$ daily closes covering January 2014 through December 2024, the same window used by the companion `JumpHMM.jl` validation paper); filtering on the common length preserved 424 tickers, of which the SPY total-return growth-rate series was taken as the market path g_m and the remaining 423 tickers as non-market assets. For each non-market ticker i we fitted a `JumpHMM` marginal to the asset's own growth-rate history, using the default configuration ($N = 100$ Laplace-partition states, Student-t emissions with $\nu = 5$, annualized growth rates via `excess_growth_rates`, risk-free rate $r_f = 0$, $\Delta t = 1/252$); jump parameters (ϵ, λ) were left at the package defaults rather than tuned per ticker, which would improve per-ticker marginal fit but was not required for a comparison that tests the effect of the composition rule on top of whatever marginal the generator produces. For each non-market ticker we then ran ordinary least squares of the asset's growth-rate history on g_m, yielding the calibrated quadruple $(\alpha_i, \beta_i, R^2_{i,\text{real}}, \sigma_{\varepsilon,\text{real},i})$. The universe spanned $\beta \in [0.01, 1.79]$ with median $\beta = 1.00$ and quartiles $(0.81, 1.00, 1.18)$, with $R^2_{i,\text{real}}$ ranging from 0.001 to 0.933 (Fig. 7).

The composition rules used a *paired innovation* protocol where applicable: for each non-market ticker and each replication $r \in \{1, \dots, R\}$ we drew a single innovation path $\tilde{\varepsilon}_i^{(r)}$ of length T from the ticker's JumpHMM marginal, and the same $\tilde{\varepsilon}_i^{(r)}$ was passed to the naive, Gaussian-residual-SIM, and hybrid composers, so that differences among those three reflect the composition rule and not generator stochasticity. The *Naive* composer set $g_i^{(r)} = \alpha_i + \beta_i\, g_m + \tilde{\varepsilon}_i^{(r)}$ with no variance correction. The *Gaussian SIM* baseline replaced the JumpHMM draw with $g_i^{(r)} = \alpha_i + \beta_i\, g_m + \sigma_{\varepsilon,\text{real},i}\, \xi_i^{(r)}$, $\xi_i^{(r)} \sim \mathcal{N}(0, I_T)$ drawn fresh per replication, recovering the calibrated R^2 exactly but discarding any marginal structure carried by $\tilde{\varepsilon}$. The *Hybrid* composer applied Algorithm 1 with floor $f = 0.10$ and threshold $R^2_{\text{preserve}} = 0.80$. We evaluated three further composers that draw their innovations from dedicated residual generators, each fit to the real OLS residual series $e_i(t) = G_i(t) - \alpha_i - \beta_i G_m(t)$. The *JumpHMM-on-residuals* composer fits a fresh JumpHMM marginal to e_i via a pseudo-price inversion (so the `JumpHMM.jl` [7] fit interface that expects a price series accepts the residual target without modification) and composes $g_i^{(r)} = \alpha_i + \beta_i g_m + \tilde{\varepsilon}^{(r)}_{\text{resid},i}$ with no variance correction. The *Block bootstrap* composer draws a Politis–Romano stationary bootstrap replicate [3] of the empirical residual series with mean block length $\sqrt{T} \approx 50$ and composes analogously. The *GARCH(1,1)-t* composer fits a conditional-variance model per ticker [2] and substitutes a simulated residual path; 30 of the 423 tickers yielded a non-stationary GARCH fit ($\alpha + \beta \geq 1$ at the optimizer's termination) and were excluded from that composer only. Across all composers, we skipped the optional cross-sectional rank reorder in Algorithm 1 so that per-ticker metrics reflected the marginal construction in isolation; cross-sectional dependence is studied separately in the companion `JumpHMM.jl` paper and is not the object of this evaluation.

For every composed series we computed three metric families against the ticker's real growth-rate history and the real SPY path: SIM recovery reported $(\hat{\alpha}_i, \hat{\beta}_i, \hat{R}_i^2)$ from OLS of $g_i^{(r)}$ on g_m along with the absolute deviations $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$; marginal fidelity was assessed by Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) two-sample p-values against the real marginal, Wasserstein-1 distance W_1, excess kurtosis κ, and the classical Hill tail-index estimator $\hat{\xi} = (k-1)^{-1} \sum_{i=1}^{k-1} \log(X_{(i)}/X_{(k)})$ on the upper 5% order statistics of $|g_i^{(r)}|$, so reported Hill values are tail indices with $\hat{\xi} \approx 1/\alpha$ for a Pareto-tailed distribution with shape α; variance preservation was the ratio $\text{Var}(g_i^{(r)})/\sigma^2_{\text{gen},i}$, using the real-data variance as a proxy for the generator's targeted variance. Pass rates were reported at $\alpha = 0.05$, and we reported $R = 100$ replications per ticker so that aggregate metrics were averages over $423 \times 100 = 42{,}300$ per-ticker observations per composer; the simulation seed was fixed at 1234.

We began with the aggregate comparison across composers, computing per-asset medians of each metric and then aggregating across the universe (Table 1). Applying the protocol to the 423 non-market tickers yielded one composed series per $(\text{ticker}, \text{composer}, \text{replication})$ triple, for a total of $423 \times 100 \times 3 = 126{,}900$ scored series. All three composers recovered β to within a median absolute error of 0.018 to 0.022, consistent with the identity $\mathrm{Cov}(g, g_m)/\sigma_m^2 \to \beta_i^{\text{eff}}$ holding at all residual scales. The differences between composers appeared everywhere else. The Gaussian SIM baseline recovered R^2 most tightly (median error 0.008) because it targeted R^2 by construction, but it failed marginal fidelity badly: KS pass rate 0.6%, AD pass rate 0.5%, excess kurtosis 1.4 (real data, ~ 13), Hill tail index 0.22 (versus ~ 0.37 on the generator; Fig. 2). The naive composition retained the generator marginal but inflated its variance by $\beta_i^2\sigma_m^2$, so its KS and AD pass rates sat at 4.5% and 2.7%. The hybrid composer held both properties at once: KS pass rate 50.4%, AD pass rate 47.6%, kurtosis 7.2, Hill 0.36; the R^2 recovery (0.021) came in tighter than the naive baseline but looser than the Gaussian baseline, reflecting that the variance-preserving branch targeted $\sigma^2_{\text{gen},i}$ rather than $R^2_{i,\text{real}}$.

We next computed the same marginal-fidelity metrics for the three residual-fit composers, and observed that all three beat hybrid on the body of the distribution (Table 1, Fig. 4). The three composers sidestep the variance-inflation problem entirely by fitting a generator directly to e_i, which by construction has variance $\sigma^2_{\varepsilon,\text{real},i}$ and no double-counted market component. Their aggregate KS pass rates were 75.8% (JumpHMM-on-residuals), 73.3% (block bootstrap), and 66.9% (GARCH(1,1)-t), each higher than hybrid's 50.4%. The advantage sat in the body of the distribution rather than the tails: hybrid's median excess kurtosis (7.17) and Hill upper-tail index (0.365) matched JumpHMM-on-residuals (7.26 and 0.365) to within sampling variability, confirming that the variance correction of Eq. (4) preserves the heavy-tailed character of the full-return generator even when the rescaling is applied. Block bootstrap and GARCH, by fitting their own residual generators, reproduced the empirical residual series but with slightly lighter tails ($\kappa = 6.68$ and 6.04; Hill 0.330 and 0.318) than either JumpHMM-based recipe.

The downstream consequence of the body-fidelity gap was the question a per-ticker one-day Value-at-Risk backtest was designed to answer (Table 5, Fig. 9). At $\alpha = 0.99$ the nominal exceedance rate is 1%; the naive composer under-breached at 0.67% and the Gaussian baseline over-breached at 1.40%, both failing Kupiec unconditional coverage on the majority of the universe (45 and 47% pass rates). The hybrid, JumpHMM-on-residuals, block bootstrap, and GARCH(1,1)-t composers all hit coverage within one-tenth of a percentage point of nominal (0.95, 0.99, 1.09, and 1.09% respectively), and their Kupiec pass rates clustered between 72 and 86%. Hybrid and JumpHMM-on-residuals were statistically indistinguishable on this test: the 25 percentage-point body-fidelity gap between them did not translate to a Value-at-Risk accuracy difference. Hybrid's standalone advantage is therefore scoped: it is the right rule when a full-return per-asset generator is the given input, and it delivers tail accuracy on par with a dedicated residual generator without requiring a separate fitting stage.

Having established that body-fidelity did not translate into VaR accuracy, we next traced where the KS advantage over the naive baseline came from by plotting per-ticker KS pass rate and per-ticker marginal variance ratio $\mathrm{Var}(g)/\sigma^2_{\text{gen}}$ as functions of the calibrated β (Fig. 1). The variance-ratio panel made the mechanism explicit: the naive scatter traced the theoretical curve $1 + \rho_i$ with $\rho_i = \beta_i^2\sigma_m^2/\sigma^2_{\text{gen},i}$, overshooting the generator variance by 50% to 75% at high β, while hybrid and Gaussian SIM sat on the unit line by design. The KS-pass-rate panel showed the consequence: naive fell from a pass rate of ~ 0.25 at $\beta \approx 0.3$ to below 0.01 past $\beta \approx 0.7$, while hybrid sat between 0.3 and 0.9 across the full β range. Gaussian SIM was pinned to zero throughout; with its Gaussian residual it did not match the heavy-tailed real marginal at any β. Having traced the KS advantage to variance preservation, we next separated the two baselines' failure modes by plotting per-ticker excess kurtosis and Hill tail index against β (Fig. 2). The naive and hybrid composers clustered together near the real-data kurtosis reference line ($\kappa_{\text{real}} \approx 13$), with the difference between them reflecting the variance inflation under naive rather than any difference in tail shape. The Gaussian SIM series collapsed to $\kappa \approx 1$ and Hill ≈ 0.22, pinned below the generator regardless of β; by replacing the heavy-tailed innovation with a Gaussian draw the baseline removed the entire heavy-tail carrier. The Hill panel confirmed the same separation at the tail: naive and hybrid both reported ~ 0.37, while Gaussian SIM reported ~ 0.22.

Having established aggregate preservation, we next examined the branch-selection outcome on this universe and asked whether the advantage held across the β distribution. Applying the branch-selection rule of Algorithm 1 to the 423-ticker universe produced 421 assignments to the `hybrid` branch, 2 to the `r2-preserve` branch, and no assignments to the `hybrid-clipped` branch (Fig. 3, Table 2). The two R^2-preserving tickers were QQQ ($R^2 = 0.861$) and SPYG ($R^2 = 0.933$), both ETFs that track market-capitalization indices closely; the highest-R^2 individual stock (BLK at $R^2 = 0.645$) sat a structural ~ 0.22 below the $R^2_{\text{preserve}} = 0.80$ threshold, so the branch assignment was empirically tracker-specific on this universe rather than a threshold artifact. The two trackers' KS pass rate under hybrid composition was 99%, consistent with the R^2-preserving branch recovering both the calibrated β and the calibrated R^2 by construction. Clipping did not trigger anywhere in the universe: the market loading ratio $\rho_i = \beta_i^2\sigma_m^2/\sigma^2_{\text{gen},i}$ stayed below $1 - f = 0.90$ for every ticker, because the JumpHMM marginals carried enough variance that the market component never crowded out the $f = 0.10$ idiosyncratic floor. Because the R^2-preserve branch had only two empirical assignments, we validated it on synthetic trackers where the target (β, R^2) was known by construction (Fig. 8). For each $(\beta_{\text{true}}, R^2_{\text{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$ we synthesized an asset via $g_i(t) = \alpha_{\text{true}} + \beta_{\text{true}} g_m(t) + \sigma_{\varepsilon,\text{true}}\,\xi(t)$ with $\sigma^2_{\varepsilon,\text{true}} = \beta^2_{\text{true}}\sigma_m^2(1 - R^2_{\text{true}})/R^2_{\text{true}}$ and ran the series through the hybrid composer at 100 replications. Across all 15 cells the branch selector assigned `r2-preserve` as intended, recovered median $\hat{\beta}$ to within 0.001 of β_{true} and median $\hat{R}^2$ to within 0.001 of R^2_{true}, and cleared the $\alpha = 0.05$ KS test on at least 99% of replications per cell. The branch is therefore correct by construction and by controlled test, with the two-ticker empirical occupancy reflecting the S&P 500 universe rather than a limitation of the composition rule. To check whether the preservation advantage held across the β distribution, we bucketed the universe by β quartile and recomputed the aggregate metrics within each bucket (Table 3). The hybrid KS pass rate declined gently from 67% in the low-β quartile to 44% in the high-β quartile; the Gaussian SIM pass rate stayed pinned near zero across all buckets; the naive pass rate collapsed from 16% in Q1 to under 1% in Q3 and Q4 as $\beta_i^2\sigma_m^2$ grew. The gentle hybrid decline at high β reflected the caveat noted alongside Algorithm 1: as $\beta^{\text{eff}\,2}\sigma_m^2$ grew toward $(1 - f)\sigma^2_{\text{gen}}$, the market-driven component of g tightened the marginal toward Gaussian along its own axis, mildly attenuating the generator's tails in the composed series. The effect was small enough that hybrid still beat both baselines by more than an order of magnitude in every bucket.

To probe the clipping branch, which did not trigger anywhere in the main run, we re-ran the hybrid composer on the same universe with the market growth-rate path rescaled by a factor $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals $(\alpha_i, \beta_i, \sigma_{\varepsilon,\text{real},i}, \sigma_{\text{gen},i})$ fixed and drawing $R = 100$ replications per (ticker, γ) pair (Fig. 5). The scale factor entered the market loading ratio as $\rho_i(\gamma) = \gamma^2\beta_i^2\sigma_m^2/\sigma^2_{\text{gen},i}$, so increasing γ simulated an environment in which market variance grew without the per-ticker generators being refit. At $\gamma = 1$ no ticker crossed the clipping threshold and the hybrid KS pass rate matched the headline value (50.3%). Doubling σ_m pushed 76.4% of tickers past $\rho_i > 1 - f = 0.90$ and the clipping branch activated accordingly; at $\gamma = 3$ the clipping rate reached 95.2%. Hybrid KS pass rates under stress were 71.8% at $\gamma = 2$ and 77.3% at $\gamma = 3$, above the unclipped baseline, because the clipping branch rescales β^{eff} to hold the idiosyncratic floor $f\sigma^2_{\text{gen},i}$ and thereby preserves marginal body shape when the market component would otherwise crowd it out. The clipping branch is therefore not a degradation under stress but a mechanism that keeps the composed marginal near the generator target as $\beta^{\text{eff}\,2}\sigma_m^2$ inflates; its cost, the downward-biased effective beta required to hold the floor, was paid only on tickers where the unclipped identity could not be maintained.

We also asked whether the hyperparameters of Algorithm 1, the R^2-preserve threshold R^2_{preserve} and the idiosyncratic floor f, had been tuned to the headline result. We re-ran the hybrid composer on the full 423-ticker universe at every point of a 5×5 grid, $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$ and $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ (Fig. 6). Hybrid KS pass rate ranged from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points, and the R^2-preserve assignment was stable at QQQ and SPYG for every threshold at or below $R^2 = 0.85$, with QQQ reassigning to the `hybrid` branch only at the $R^2_{\text{preserve}} = 0.90$ corner. The floor f did not change any branch assignment in the grid, since the maximum $\rho_i = \beta_i^2\sigma_m^2/\sigma^2_{\text{gen},i}$ in the universe stayed below $1 - f$ for every $f \in \{0.05, \ldots, 0.30\}$. The headline number is therefore invariant to the hyperparameter values within wide tolerances, and the operating point ($R^2_{\text{preserve}} = 0.80,\ f = 0.10$) reflects a principled choice of the R^2-branch threshold and a standard floor value rather than a data-peeked local optimum.

Finally, we quantified the contribution of generator stochasticity to the headline numbers by re-running the full six-composer protocol under 8 random seeds (Table 4). The per-composer KS pass rates held to within fractions of a percentage point: hybrid sat at $50.40 \pm 0.18\%$ across seeds, JumpHMM-on-residuals at $75.61 \pm 0.11\%$, block bootstrap at $73.24 \pm 0.09\%$, GARCH(1,1)-t at $66.96 \pm 0.30\%$, naive at $4.41 \pm 0.07\%$, and Gaussian SIM at $0.60 \pm 0.04\%$. The across-composer differences therefore exceeded the seed-induced spread by roughly two orders of magnitude. The seed sweep also reproduced the relative ordering of the composers without inversions across seeds, confirming that the hybrid-vs-baseline gap and the residual-fit-vs-hybrid body-fidelity gap were not artifacts of the simulation seed.

4 Discussion

In this study, we composed synthetic per-asset growth-rate paths under a single-index model (SIM) by drawing each asset's innovation from an independent JumpHMM generator fit to the asset's own full growth-rate history, then rescaling the draw by a closed-form scalar before inserting it into the SIM composition. We evaluated six composers on a calibrated universe of 424 US equities and exchange-traded funds against the real SPY market path: the naive composition, a Gaussian-residual SIM baseline, the hybrid construction proposed here, and three residual-fit alternatives (JumpHMM-on-residuals, Politis–Romano block bootstrap [3], and a per-ticker GARCH(1,1)-t fit [2]). The one-line variance correction, together with the sign-preserving clip rule and the R^2-preserving branch, closes a gap between two previously incomplete options for building synthetic asset paths with both marginal structure and factor structure: naive composition gives both but distorts the marginal by $\beta^2\sigma_m^2$, while Gaussian-residual SIM gives the factor exactly but strips the marginal. The hybrid construction sat between the Gaussian-residual and naive baselines: it recovered β to the same precision as either, held the generator marginal to its target variance in every ticker, and preserved the generator's tails at a level indistinguishable from the naive composition (Fig. 1, Table 1). The three residual-fit composers produced higher Kolmogorov–Smirnov pass rates than hybrid on marginal body fidelity (67 to 76% versus 50%, Table 1), but matched hybrid on the 99% Value-at-Risk coverage test to within statistical tolerance (Table 5); the body-fidelity gap did not translate to a tail-accuracy gap. The contribution of hybrid was therefore scoped to the case where the per-asset generator was specified in full-return units (a common setting, since per-asset generators are often calibrated for distributional interpretability, cached for other pipelines, or chosen to retain regime-switching or jump structure); in that case hybrid delivered 99% VaR accuracy on par with a dedicated residual generator without requiring a separate fitting stage. The variance-inflation gap was large enough to matter in practice: at the high-β end of the universe, naive composition inflated the marginal variance by over 70% relative to the generator target (Fig. 1). Because branch selection was driven entirely by the calibrated R^2 and each asset carried a persisted construction flag, the procedure ran uniformly across the ticker universe without external metadata and left an audit trail that downstream consumers could inspect. No part of the construction is new in a deep sense; the contribution is a minimal, practitioner-ready recipe with an auditable branch flag and a principled treatment of the degenerate tracker limit, usable as a drop-in replacement for naive composition in workflows where both tail shape and market dependence matter.

The evaluation isolated the composition rule by fixing the market path at the real SPY history and using the same per-ticker innovation draw across all three composers. This design yielded a paired comparison of composition rules at the per-ticker level but set aside two questions that production users will need to address separately. The first concerns cross-sectional dependence: we skipped the optional rank-reorder step in Algorithm 1 so that per-ticker metrics reflected the composition rule alone. When the reorder is applied, the variance algebra is unchanged (rank-reordering preserves the marginal variance of ε_i), so the marginal results here carry over; cross-sectional fidelity under different copula specifications is studied in the companion `JumpHMM.jl` paper [7]. The second concerns market-path uncertainty: the metrics reported here used the real SPY path, and under a simulated market path the composed marginal picks up additional variance from the sampling of $g_m^{(r)}$, which for high-R^2 tickers becomes a noticeable effect. In production, the two deferred questions compose: a reorder-plus-simulated-market pipeline would layer cross-sectional structure on top of per-ticker draws built against sampled market paths, with the variance correction of Eq. (4) still providing the per-asset variance target on each realized $g_m^{(r)}$. A separate scope boundary concerns the downstream use case: the main metrics reported here characterize marginal and factor fidelity of the composition rule on per-ticker targets. To test whether that fidelity translates to a consequential downstream number, we also report a per-ticker one-day 95% and 99% Value-at-Risk backtest of the composed paths against the real 2014–2024 growth-rate histories, with exceedance counts assessed by Kupiec's unconditional coverage test; the per-ticker framing keeps this validation inside the paper's scope, while portfolio-level VaR requires the cross-sectional copula deferred to the companion paper [7]. High-turnover portfolio strategies remain a distinct scope boundary: the temporal structure of the residual draw becomes load-bearing in a way that contemporaneous composition does not address.

A related boundary on the recipe was the clipping rule. Clipping did not trigger anywhere in the US-equity universe used here, because the JumpHMM marginals carried enough variance that $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stayed below 0.90 even for the highest-β ticker. Clipping becomes relevant in two settings we did not evaluate: scenario-driven markets, where the synthetic g_m is constructed to be much more volatile than the calibration market (e.g., stress tests), and universes that mix low-volatility single stocks with extreme betas (e.g., leveraged ETFs). In both settings we recommend reporting the original and effective β_i alongside the output, as in Algorithm 1, so that downstream consumers can audit the attenuation. A related attenuation appeared at the other end of the β spectrum, even when clipping did not trigger: the hybrid KS pass rate fell from 67% at low β to 44% at high β (Table 3), a mild attenuation that followed directly from the composition algebra. As $\beta_i^2 \sigma_m^2$ grows toward the idiosyncratic floor $(1-f)\sigma_{\text{gen},i}^2$, the market-driven component carries an increasing share of the marginal, and that component is only as heavy-tailed as g_m. The market's own heavy tails slow the attenuation (SPY growth rates are themselves heavy-tailed after JumpHMM fitting), but the effect is visible. Users who need exact marginal fidelity at high β can recover it by lowering the floor f (at the cost of less conservative clipping behavior) or by pre-scaling the generator variance so that ρ_i stays small.

The variance correction $s_i^2 = 1 - \rho_i$ relies on $\mathrm{Cov}(\tilde{\varepsilon}_i, g_m) \approx 0$. The JumpHMM marginals are fit in isolation and sampled independently of the market path, so this holds by construction; if the generator is already correlated with the market (for example, a bootstrap from joint history), the empirical covariance should be subtracted from β_i before applying the correction, or s_i^2 should be estimated directly from the empirical variance of the composed series. The same warning applies to the R^2-preserving branch, whose variance identity assumes ε is uncorrelated with g_m after any reorder. The construction also injects a contemporaneous market factor only: $g_i(t)$ depends on $g_m(t)$ and on $\tilde{\varepsilon}_i(t)$ but not on $g_m(t-1), g_m(t-2), \ldots$, so lagged market dependence (lead-lag effects, market-driven volatility clustering, asymmetric responses to drawdowns) is not reproduced by this recipe. The natural extension is a lagged-factor SIM, $g_i(t) = \alpha_i + \sum_{k=0}^{K} \beta_{i,k}\, g_m(t-k) + \varepsilon_i(t)$; the variance correction generalizes to that form by replacing $\beta_i^2 \sigma_m^2$ with the variance of the full market-factor component, though the empirical value of the extra parameters is beyond the scope here. Lagged idiosyncratic dependence is likewise absent: the JumpHMM residuals $\tilde{\varepsilon}_i$ are drawn independently across time, so any residual autocorrelation present in the real data is discarded by the composition. This matters most for workflows that evaluate portfolios rebalancing at high frequency. With iid residuals, a rebalancing rule collects the full diversification return of [8], a compounding premium that scales with residual variance and rebalancing frequency and is largely absent in real markets, where daily residual autocorrelation is small but positive [9]. Two generator alternatives in Table 1 preserve temporal structure directly: the block-bootstrap composer resamples blocks of the empirical OLS residual series [3], inheriting whatever autocorrelation is present in the data, and the GARCH(1,1)-t composer fits a conditional-variance model per ticker [2]. Both recover marginal fidelity at the cost of committing the pipeline to a specific residual generator, whereas the variance correction of Eq. (4) is generator-agnostic and accepts a full-return JumpHMM marginal (with its jump and regime structure) as input. Users evaluating high-turnover strategies on JumpHMM-corrected paths should benchmark absolute return levels against a static-allocation baseline on the same paths, or adopt the bootstrap or GARCH alternatives when residual autocorrelation is the load-bearing structure for the downstream task.

The diversification-return artifact is quantitatively large enough on standard rebalancing rules that absolute-return claims drawn from JumpHMM-corrected paths require an explicit correction or an explicit caveat. As a concrete demonstration, a 20-asset target-weight allocator deployed against a hybrid-composed 2025 universe (calibration window 2014–2024) produced a median synthetic continuously-compounded growth rate (CCGR) of $\sim 30\%/\mathrm{yr}$, while the same allocator deployed against the real 2025 growth-rate history produced $11.5\%/\mathrm{yr}$. The decomposition isolated the artifact: equal-weight buy-and-hold on the synthetic ensemble produced $\sim 7.7\%/\mathrm{yr}$, the same target-weight allocation held statically (no rebalance) produced $\sim 2.0\%/\mathrm{yr}$, and switching the same allocation rule to daily rebalancing produced $\sim 29.7\%/\mathrm{yr}$, a $+27.7$ percentage-point jump from the static-to-daily switch on the same allocation rule and the same paths (Online Appendix S2, Table **??**). The ~ 22 pp gap between the synthetic and real engine CCGR was therefore a structural feature of the iid-residual generator family rather than a calibration error, consistent with [8–10].

Two practical responses follow. Where the workflow can tolerate being explicit about the issue, we recommend reporting the static-versus-rebalanced decomposition on the same allocation rule and the same synthetic paths, and framing absolute Sharpe levels under daily rebalancing as "this generator with this strategy" rather than as a forward prediction. Where the workflow requires an absolute-return level for downstream consumption (e.g., percentile diagnostics overlaying real outcomes against a synthetic distribution), we recommend a uniform location-shift bias correction that anchors the corrected synthetic median CCGR to a documented long-run prior:

$$W_{p,t}^{\mathrm{corr}} \;=\; W_{p,t} \cdot \exp\!\Big(-\,\tfrac{b}{100}\,\cdot t\,\Delta t\Big), \qquad b \;=\; \operatorname*{median}_{p} \mathrm{CCGR}_p \;-\; \mathrm{CCGR}^{\mathrm{prior}}, \tag{10}$$

where $W_{p,t}$ is path-p wealth at integer step t, Δt is the integration step in years, and $\mathrm{CCGR}^{\mathrm{prior}}$ is set from an external long-run prior (typically near $8\%/\mathrm{yr}$, decomposable into the prevailing risk-free rate plus an active premium net of frictions [11, 12]). The correction is a per-path monotone transform: it preserves the rank order of paths by terminal wealth, the drawdown shape, and the volatility envelope; only the level shifts. Anchoring to a documented prior rather than to the realized test-year outcome keeps the correction independent of the sample to which the synthetic ensemble is being compared and avoids circular calibration. The realized year then lands inside the corrected ensemble at an honest percentile (in our deployment, real-2025 landed at the 63rd percentile of the corrected ensemble, consistent with 2025 being an above-typical equity year). Online Appendix S2 details the methodology, contrasts the location-shift correction with a multiplicative haircut alternative (which we do not recommend because it scrambles per-path rank order and distorts the drawdown distribution), and discusses why several intuitive alternatives (parameter drift between calibration and deployment, transaction-cost inflation, AR(1) on residuals without restructuring the copula sampling, residual block-bootstrap) do not address the underlying mechanism within the iid-residual SIM framework. The synthetic data remained reliable, both before and after correction, for the ensemble-shape questions the paper validates: percentile structure, drawdown distribution, and relative ranking of strategies on the same paths. The location-shift correction extends that reliability to absolute Sharpe and CCGR levels under daily rebalancing, at the cost of an external prior.

Two further extensions follow naturally. Multi-factor SIM replaces the single market factor with K factors, requiring ρ_i to become the sum of factor contributions and the loading vector to be clipped rather than a scalar, though the branch logic is unchanged. Per-ticker tuning of the JumpHMM jump parameters (ϵ, λ) tightens each marginal's fit; we used defaults to keep fitting tractable across the universe, but tuning as reported in the companion paper [7] would push the hybrid KS pass rates upward. A stress-test harness ties these together: using the clipping rule to enforce a minimum idiosyncratic share in constructed market scenarios guarantees that per-asset marginals remain heavy-tailed even under synthetic markets much more volatile than history, which is precisely the regime where naive SIM composition breaks down most severely.

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Author contributions. A.A. designed the hybrid hidden Markov marginal generator, performed the single-asset experiments and validation framework, and contributed to the manuscript. J.D.V. designed the variance-corrected single-index composition, performed the multi-asset experiments and downstream Value-at-Risk validation, developed the bias-correction methodology, released the code, and contributed to the manuscript.

Competing interests. The authors declare no competing interests.

Code and data availability. All source code, experiment scripts, and fitted per-asset marginals are released as part of `JumpHMM.jl` (https://github.com/varnerlab/JumpHMM.jl); the paper-specific evaluation harness (six-composer benchmark, Value-at-Risk backtest, bias-correction utility) lives in the `paper/` subdirectory of that repository.

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Table 1: **Aggregate scorecard across** 423 **non-market tickers and** 100 **replications per ticker.** Median values per composer. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$. All three composers recover β to the same precision; only the hybrid composer preserves the marginal distribution (KS pass 50.4%) while holding the variance close to the generator target.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365
JumpHMM-on-residuals	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.017	0.036	67.4	61.4	0.267	6.069	0.317

Table 2: **Hybrid composer broken out by construction flag.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The `hybrid-clipped` branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the `r2-preserve` branch (QQQ and SPYG) attain a KS pass rate of 99%, consistent with the branch recovering both β and R^2 by construction.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	50.1	0.273	7.142
`r2-preserve`	2	0.005	0.001	99.0	0.109	11.525

Table 3: **Aggregate metrics by** β **quartile, across composers.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines gently from 67% in Q1 to 44% in Q4 as the market loading ratio ρ_i grows, consistent with the tail-attenuation caveat of Section 2.6; naive collapses from 16% to under 1% over the same range.

β bucket	Composer	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	15.7	0.414	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	67.3	0.200	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)-t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	0.9	0.632	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	45.9	0.254	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)-t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.4	0.802	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	44.6	0.297	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)-t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	0.8	0.976	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	43.7	0.363	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)-t	0.021	64.5	0.361	6.064	34.7

Table 4: **Seed-uncertainty summary across 8 random seeds.** Each cell reports mean $\pm$ SD across the 8 seeds in $\{1234, 2345, 3456, 4567, 5678, 6789, 7890, 8910\}$, with each seed running the full 423-ticker $\times$ 100-replication protocol. Conditional-variance paths for GARCH(1,1)-t are re-simulated at scoring time so its seed SD is informative. The between-composer differences in KS pass rate exceed the seed-induced SD by roughly two orders of magnitude (hybrid SD 0.18 pp vs 25 pp gap to JumpHMM-on-residuals).

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\mathrm{real}}\rvert$	KS pass (%)	AD pass (%)	κ	Hill
Naive	0.022 ± 0.000	0.088 ± 0.000	4.41 ± 0.07	2.71 ± 0.02	7.746 ± 0.015	0.369 ± 0.000
Gaussian SIM	0.017 ± 0.000	0.008 ± 0.000	0.60 ± 0.04	0.50 ± 0.01	1.428 ± 0.005	0.217 ± 0.000
Hybrid	0.018 ± 0.000	0.021 ± 0.000	50.40 ± 0.18	47.50 ± 0.19	7.150 ± 0.011	0.364 ± 0.000
JumpHMM-on-residuals	0.018 ± 0.000	0.015 ± 0.000	75.61 ± 0.11	70.51 ± 0.12	7.250 ± 0.013	0.365 ± 0.000
Block bootstrap	0.017 ± 0.000	0.020 ± 0.000	73.24 ± 0.09	66.79 ± 0.11	6.676 ± 0.011	0.330 ± 0.000
GARCH(1,1)-t	0.017 ± 0.000	0.036 ± 0.000	66.96 ± 0.30	61.00 ± 0.26	6.072 ± 0.022	0.318 ± 0.000

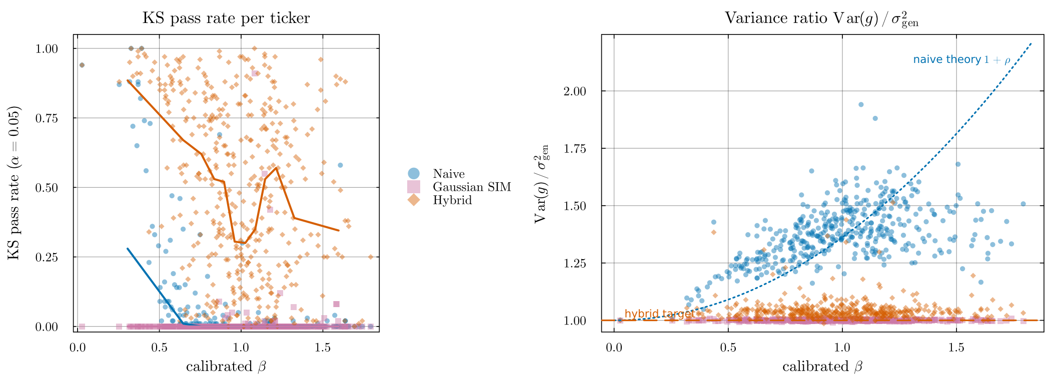

Figure 1: **Marginal-distribution preservation and variance preservation vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$. The hybrid composer (orange) maintains a pass rate between ~ 0.3 and ~ 0.9 across the full β range; the naive composer (blue) collapses past $\beta \approx 0.7$ as the market loading inflates the marginal variance; the Gaussian SIM baseline (purple) is pinned at zero throughout. *Right:* variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ^2_{gen}; the naive scatter follows it. Hybrid and Gaussian SIM sit on the unit line by design.

Table 5: **Per-ticker one-day Value-at-Risk backtest across the six composers.** For each (ticker, replication) pair we estimate the VaR threshold at coverage level α from the composed path and count exceedances against the real 2014–2024 growth-rate history. *rate*: mean exceedance rate across tickers (nominal: $1-\alpha$). *Kupiec pass*: fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals are statistically indistinguishable at both coverage levels; both strawman composers (naive, Gaussian SIM) fail coverage on a majority of the universe.

	$\alpha = 0.95$		$\alpha = 0.99$	
Composer	rate (%)	Kupiec pass (%)	rate (%)	Kupiec pass (%)
Naive	3.55	10.4	0.67	45.2
Gaussian SIM	4.06	42.6	1.40	47.0
Hybrid	4.75	80.4	0.95	79.7
JumpHMM-on-residuals	4.87	84.0	0.99	82.9
Block bootstrap	4.78	76.7	1.09	86.0
GARCH(1,1)-t	4.64	69.0	1.09	72.1

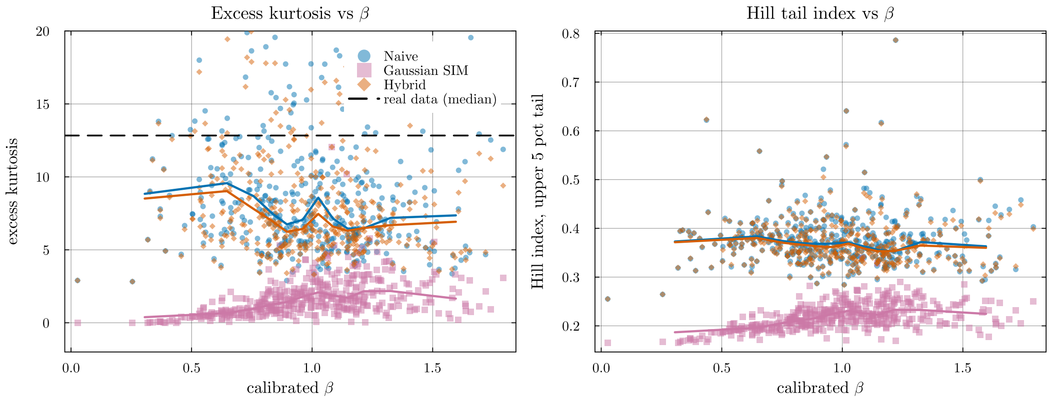

Figure 2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\mathrm{real}} \approx 13$); the hybrid and naive composers (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians sit near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22, indicating the shift toward a thinner-tailed distribution. The hybrid's median sitting below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion; the panel's point is that hybrid tracks naive (inheriting generator tails) while Gaussian SIM does not.

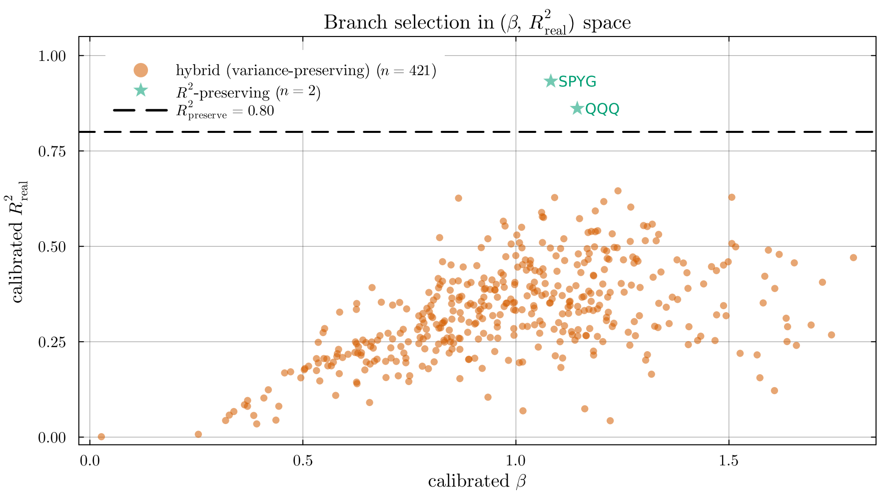

Figure 3: **Construction-flag assignment in calibrated $(\beta, R^2_{\mathrm{real}})$ space.** The $R^2_{\mathrm{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from genuine single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the `r2-preserve` branch. The remaining 421 tickers populate the `hybrid` (variance-preserving) branch; the `hybrid-clipped` branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\mathrm{gen},i}^2$ stays below 0.90 for every ticker.

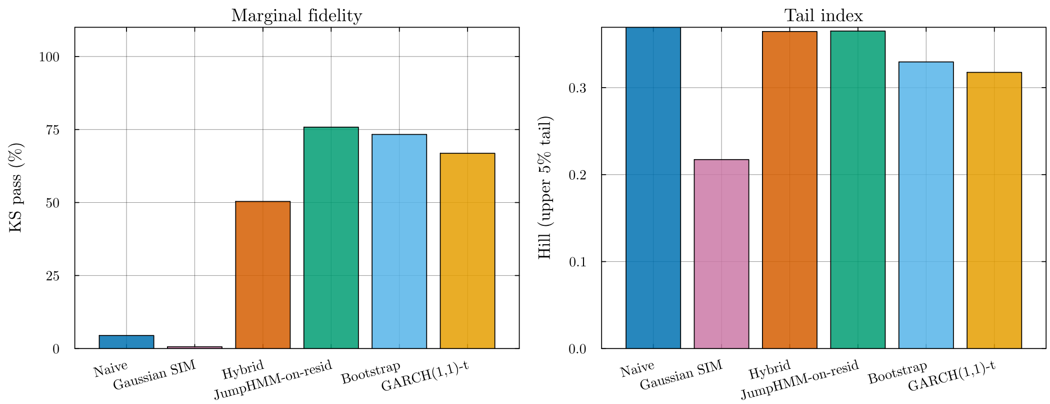

Figure 4: **Six-composer comparison on aggregate marginal fidelity and tail index.** *Left:* per-ticker Kolmogorov–Smirnov pass rate at $\alpha = 0.05$, aggregated across 423 tickers and 100 replications per ticker. The three residual-fit composers (JumpHMM-on-residuals, block bootstrap, GARCH(1,1)-t) dominate the hybrid on marginal body fidelity; both strawman composers collapse. *Right:* median Hill tail index on the upper 5% of $|g|$. Hybrid, JumpHMM-on-residuals, and naive cluster at ~ 0.37, reflecting the shared full-return JumpHMM generator's tail character; block bootstrap and GARCH sit slightly lower at 0.33–0.32; Gaussian SIM collapses to 0.22.

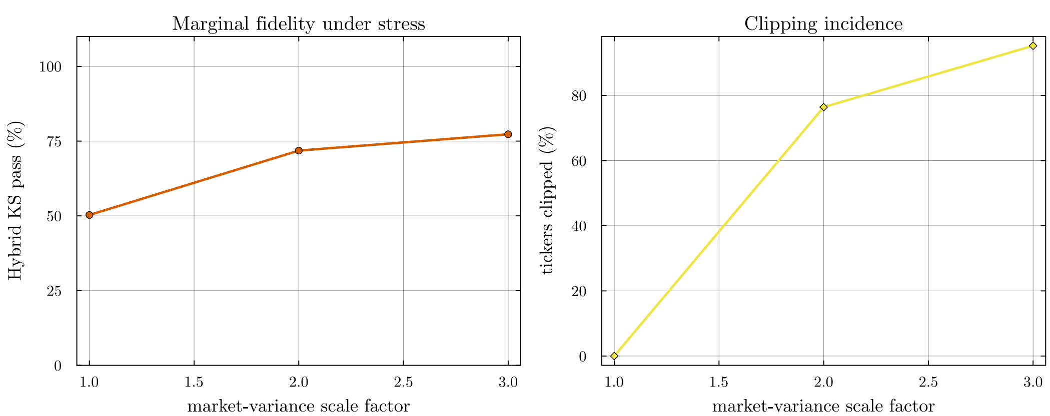

Figure 5: **Clipping-branch behavior under market-variance stress.** Hybrid composer re-run with the SPY growth-rate path rescaled by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed. *Left:* hybrid KS pass rate at $\alpha = 0.05$ rises from the baseline 50.3% at $\gamma = 1$ to 71.8% and 77.3% at $\gamma = 2$ and $\gamma = 3$, because the clipping branch rescales β^{eff} to hold the idiosyncratic floor and prevents the market component from crowding out the generator's marginal. *Right:* fraction of tickers for which the clipping branch activated ($\rho_i > 1 - f = 0.90$). No ticker triggered clipping in the main run; doubling σ_m pushed 76% of the universe past the threshold, and $3\times$ pushed 95%. The branch is a robustness mechanism, not a degradation mode.

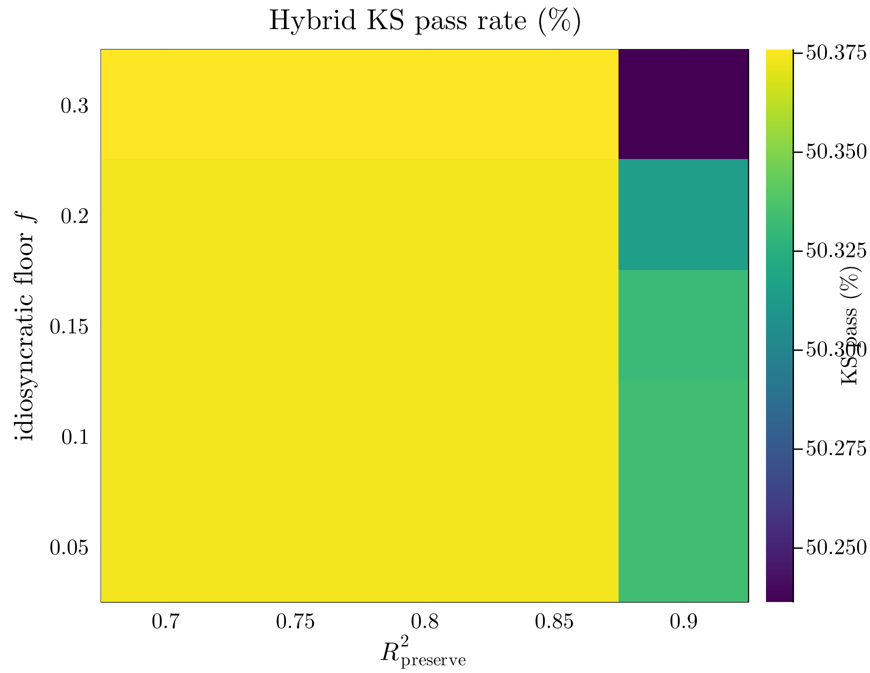

Figure 6: **Hybrid KS pass rate across a 5×5 hyperparameter grid.** Columns index the R^2-preserve threshold $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$; rows index the idiosyncratic floor $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Hybrid KS pass rate varies from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points. The operating point $(0.80, 0.10)$ used throughout the paper is indistinguishable from every other grid point within sampling noise, indicating that the headline result is not a data-peeked local optimum.

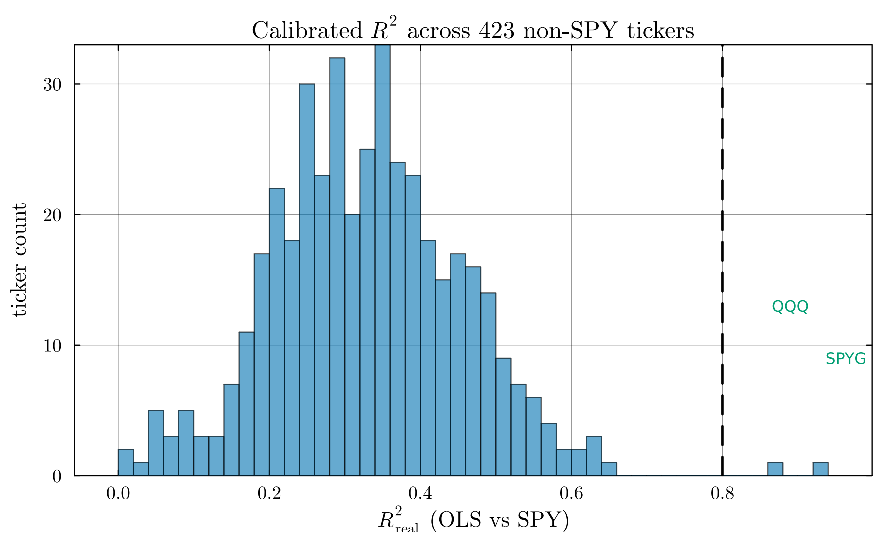

Figure 7: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of OLS R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index ETFs (QQQ, SPYG); the highest-R^2 individual stock (BLK, 0.645) sits a structural ~ 0.22 below the threshold. The `r2-preserve` branch's $n = 2$ occupancy is therefore a property of the single-factor SIM on S&P 500 names against SPY, not a sample-size artifact or a data-peeked threshold.

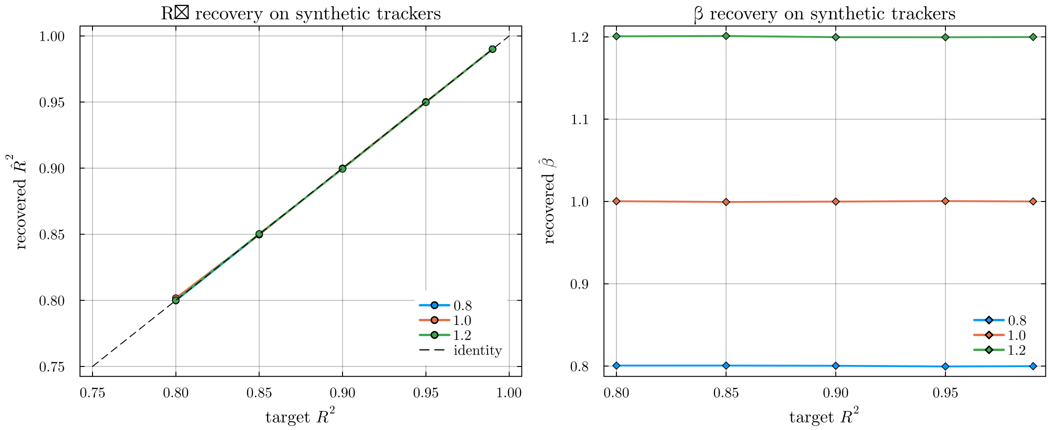

Figure 8: R^2**-preserve branch validated on synthetic trackers.** For each $(\beta_{\text{true}}, R^2_{\text{true}})$ with $\beta_{\text{true}} \in \{0.8, 1.0, 1.2\}$ and $R^2_{\text{true}} \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$, a synthetic tracker was constructed from $\sigma^2_{\varepsilon,\text{true}} = \beta^2_{\text{true}}\sigma^2_m(1 - R^2_{\text{true}})/R^2_{\text{true}}$ and composed via Algorithm 1 at 100 replications. *Left:* recovered $\hat{R}^2$ against target R^2, grouped by β_{true}; the identity line is dashed. *Right:* recovered $\hat{\beta}$ against target R^2. Across all 15 cells the branch recovered median $\hat{\beta}$ to within 0.001 of β_{true}, median $\hat{R}^2$ to within 0.001 of R^2_{true}, and cleared the $\alpha = 0.05$ KS test on at least 99% of replications per cell.

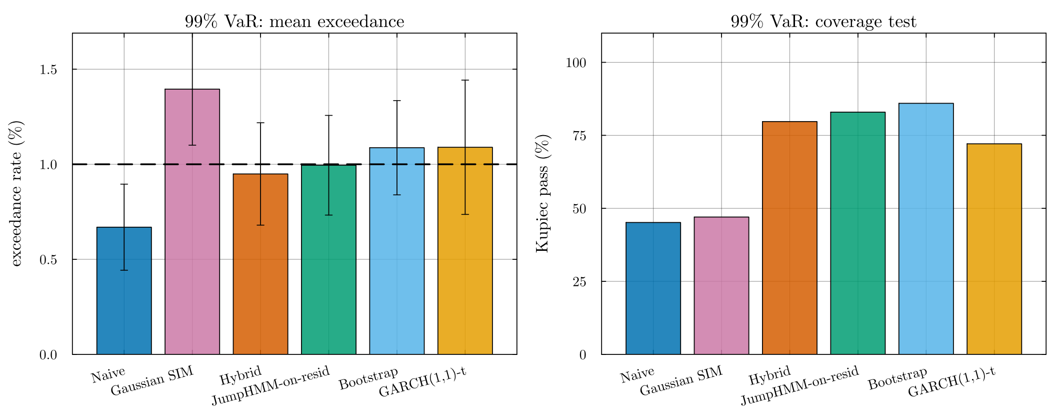

Figure 9: 99% **Value-at-Risk exceedance rates and Kupiec coverage across the six composers.** *Left:* mean exceedance rate across tickers (nominal 1%, dashed line); error bars are one SD across tickers. Hybrid, JumpHMM-on-residuals, block bootstrap, and GARCH(1,1)-t all hit coverage within 0.1pp of nominal; naive under-breaches (inflated variance in the composed marginal) and Gaussian over-breaches (thin tails). *Right:* fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals are statistically indistinguishable (80 and 83%); the strawman composers fail on the majority of the universe.

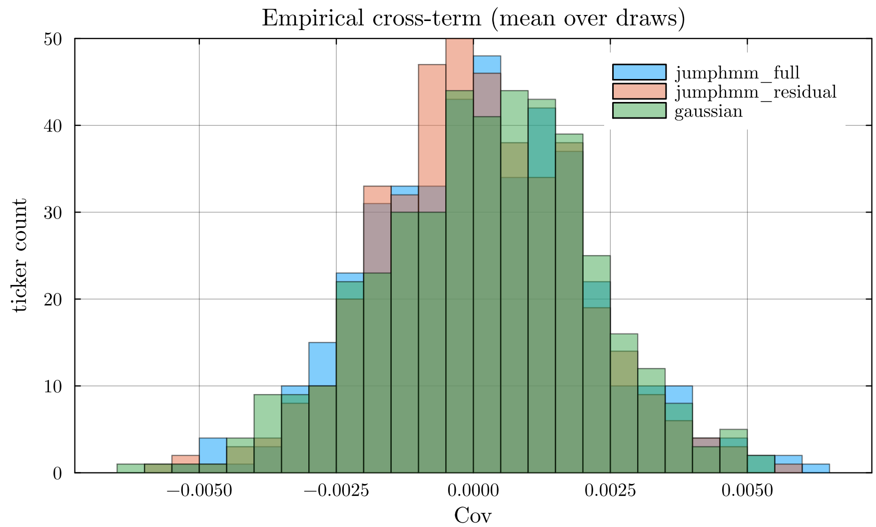

Figure S1: **Empirical normalized cross-covariance** $\mathrm{Cov}[\tilde{\varepsilon}_i, g_m]/(\sigma_{\tilde{\varepsilon},i}\,\sigma_m)$ **across the** 423**-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ innovation draws. The full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators all clustered at zero, validating the independence assumption used to derive Eq. (4).

A Empirical cross-term diagnostic

The variance correction of Eq. (4) rests on the assumption $\mathrm{Cov}[\tilde{\varepsilon}_i, g_m] \approx 0$, which is what allows the derivation to drop the cross-term in the expansion of $\mathrm{Var}[g_i]$. Because the per-asset JumpHMM marginals are fit to each asset's full growth-rate history, including its market-correlated component, the assumption is not automatic: a generator that picked up a residual market loading would invalidate the closed-form scaling. We tested the assumption empirically by sampling $R = 100$ innovation paths $\tilde{\varepsilon}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real SPY path:

$$\mathrm{cov}_i^{\mathrm{norm}} = \frac{1}{R}\sum_{r=1}^{R}\frac{\mathrm{Cov}[\tilde{\varepsilon}_i^{(r)},\, g_m]}{\sigma_{\tilde{\varepsilon},i}\,\sigma_m}. \tag{S1}$$

Per-ticker values were plotted as histograms in Fig. S1. Across the 423 tickers, the full-return JumpHMM generator (`jumphmm_full`, used by hybrid and naive) sat within ± 0.006 of zero with a per-ticker mean bias of 1.5×10^{-4} (Fig. S1); the residual-fit JumpHMM generator (`jumphmm_residual`, used by JumpHMM-on-residuals) and the Gaussian baseline were also centered at zero (mean 7×10^{-5} and 2.3×10^{-4} respectively, Fig. S1). The independence assumption was therefore satisfied to within sampling tolerance for all three generators, and the derivation of Eq. (4) held in practice.

B Bias correction for daily-rebalanced backtests

S2.1 The diversification-return artifact under iid SIM residuals

S2.2 Empirical demonstration on a 20-asset deployed allocator

S2.3 Location-shift bias correction

S2.4 Why alternatives don't work (or have worse trade-offs)

S2.5 Validation and downstream-use guidance

S2.6 Reproducibility